ABHIMANYU DUDEJA

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EDUCATION

Northeastern University, Boston, MA — Expected Dec 2027
Master of Science in Computer Science
Relevant Coursework: Generative AI, Algorithms, Database Systems, Program Design Paradigms

SRM Institute of Science and Technology, Chennai, India — Jun 2025
Bachelor of Technology in Computer Science

TECHNICAL KNOWLEDGE

Languages	Python, SQL, R, Java, C, C++
ML/DL	TensorFlow, Keras, Scikit-learn, OpenCV, PyTorch
Data	NumPy, Pandas, Matplotlib, Seaborn, Plotly, Power BI
Databases	MySQL, PostgreSQL, MongoDB
Tools	Jupyter, Git, AWS (Glue, S3), Oracle Cloud
Certifications	Machine Learning (Coursera), ML Foundations (AWS), Google Data Analytics, Oracle Cloud Infrastructure, Python Development

WORK EXPERIENCE

Mob, Boston, MA — Sep 2026 - Present
Full Stack Software Engineer (Co-op)
- Delivering features end-to-end across the consumer mobile app and B2B event organizer dashboard, working directly with the founding team.

Deloitte, Remote — Jul 2023 - Aug 2023
Data Engineering Intern
- Improved 40% data ingestion speed by optimizing AWS-based ETL pipeline using SQL and AWS Glue for large-scale analytical datasets.
- Developed automated dashboards and monitoring scripts for real-time pipeline health tracking and anomaly detection.

Tech Mahindra, Noida, India — Jun 2023 - Jul 2023
Data Analyst Intern
- Cut 50% manual reporting by automating KPI report generation using Python and SQL.
- Created interactive dashboards with drill-down capabilities enabling data-driven decision-making for stakeholders.

PROJECTS

Grounded RAG Fact-Checker, U.S. Immigration & Tax Law — Jun 2026 - Aug 2026
- Indexed 14,726 chunks of statutory text into a local RAG fact-checker using FAISS, BAAI/bge-small-en-v1.5 embeddings, reciprocal rank fusion, and a MiniLM cross-encoder reranker.
- Benchmarked 4 pipeline configurations on 73 hand-labeled claims, isolating retrieval (R@8 0.769, MRR 0.375) from verdict accuracy (0.534); claim decomposition cost 9 accuracy points at 8B scale.

2026 FIFA World Cup Prediction System — May 2026 - Jul 2026
- Trained 49,378-match Elo and bivariate Poisson models with a Transfermarkt squad-value adjustment, reaching 0.890 test log loss against a 1.043 baseline (14.7% improvement).
- Scored 62.5% outcome accuracy over 72 group matches from predictions locked before kickoff, correctly calling Spain as champion at 0.893 live log loss, within 0.003 of held-out test loss.

Multi-Stage Brain Tumor Classification using DNNs — Jan 2025 - May 2025
- Achieved 97% test accuracy by developing a CNN model using TensorFlow/Keras for multi-stage brain tumor classification on medical imaging datasets.
- Curbed 40% overfitting by applying batch normalization, dropout layers, and data augmentation techniques across training pipelines.

PUBLICATIONS

- *Multi-Stage Brain Tumor Classification Using Deep Neural Networks*, ICRTC 2025 (Springer), LNNS vol. 1906, pp. 176–185.
- *Automatic Detection and Classification of Brain Tumors Using Deep Learning Model*, ICNSBT 2025 (Springer), LNNS vol. 1405, pp. 451–465.